

Kadhirvel M

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Objective

Research and Engineering Scientist with expertise in **independent research, system design, and emerging technologies**. Skilled in building autonomous AI systems, orchestrating multi-agent pipelines, and deploying scalable **LLM** and generative AI solutions in **cloud computing** and production environments. Experienced in **technical documentation, drafting specifications, and collaborative product development**, seeking a **Research and Engineering Scientist** role at **Zoho** to contribute to innovative AI, computing, and system design research. **Additionally, proficient in compiler design and its application in AI systems.**

Education

Manakula Vinayagar Institute of Technology [🔗](#)

Sep 2022 – May 2026

BTech in Artificial Intelligence & Machine Learning

CGPA: 8.42/10

- **Coursework:** Deep Learning, NLP, Computer Vision, Transformers, Multimodal Learning, AI Ethics, Algorithms, Computer Architecture, Data Structures, Software Engineering, and Computer Science.

Experience

Agentic AI Research Intern

Chennai, India

Tata Consultancy Services (TCS) [🔗](#)

June 2025 – Aug 2025

- Conducted **independent research** and **novelty identification** in multi-agent AI systems for automated package migration across cloud platforms.
- Designed and implemented **system architecture, technical specifications, and AI pipelines** using **Autogen, MCP, Python, Azure, and GCP**. Collaborated with support teams to enhance system supportability.
- Deployed **LLMs** and **generative models** in production via **Docker and Kubernetes**, ensuring scalability, reliability, and maintainability.
- Generated and augmented synthetic data, performed **data cleaning and parsing**, and optimized model inputs for **machine learning, NLP, and image processing tasks**.
- Drafted detailed **technical documentation, reports, and presentations** for internal and external communication, including miscellaneous insights on AI topics.

Technical Developer

Puducherry, India

Ignite Skylabs [🔗](#)

Mar 2024 – May 2024

- Built and deployed full-stack AI applications, improving system efficiency and client satisfaction.
- Conducted team workshops and mentoring in Tamil, emphasizing collaboration, task prioritization, and knowledge sharing in web technologies.
- Produced project documentation and specifications, supporting long-term maintenance and product lifecycle.

Skills

Computer & Engineering Systems::

- Computer Architecture, Computer Networking, Data Structures, Software Engineering, Programming Languages, Image Processing, Robotics, Automation

AI & Multimodal Modeling:

- Autogen, MCP, Transformers (GPT, BERT, RoBERTa), GANs, VAEs, LLaMA, PyTorch, TensorFlow

Cloud & Containerizations:

- AWS, Azure, GCP, Kubernetes, Docker, Production Environments

NLP & Text Processing:

- Text Cleaning, Parsing, Sentiment Analysis, Text Generation, Chunking, Embeddings

Data Management & Synthetic Data:

- Synthetic Data Generation, Data Labeling, Augmentation, NLP Pipelines, Datasets

AI Pipelines & Deployment:

- Modeling, Cloud Deployment, Optimization, Monitoring, DevOps

Soft Skills:

- Analytical, Innovative, Responsible, Collaborative, Problem-Solving, Independent, Attention to Detail, Organization, Task Prioritization, Communication, Supportive

Publications

IEEE Xplore: WiFi-based Attendance Detection	Dec 2024
Conference: AI-Powered Neurodiagnostics	Apr 2024
IEEE Xplore: Fake News Detection on Social Media	Nov 2023

Projects

Autonomous Package Migrator (TCS)

- Multi-agent GenAI system for codebase translation across stacks, using Autogen, MCP, and Python.
- Developed system design, technical specifications, and deployment architecture for containerized microservices via Kubernetes on Azure.

Alzheimer's & Parkinson's - Neurodegenerative Disease prediction

- Built EfficientNet B7 with autoencoders and attention modules, improving diagnosis accuracy by 40%.
- Deployed models via AWS SageMaker, demonstrating machine vision, image processing, and biomedical AI research.

Agentic RAG Database Chatbot

- Designed a Retrieval-Augmented Generation (RAG) chatbot integrated with a knowledge database for dynamic query responses.
- Implemented using Autogen, MCP, and Python, with multi-agent orchestration for context-aware responses.
- Developed system design, technical specifications, and backend architecture, deployed on cloud infrastructure for scalable production use.

Certifications

Data Analytics - LinkedIn 	Sept, 2021
Foundations: Data, Data, Everywhere - Coursera 	Oct, 2021
Google Project Management - Coursera 	Mar, 2022
Big Data in the Age of AI - LinkedIn 	Sept, 2021
Development & Advanced Engineering - Accenture 	Nov, 2024

Achievements

- **Winner** – *Castilo'25 Tech Hackathon*, March 2025
- **Runner-up** – *SCIMIT'25 Project Expo*, February 2025
- **Runner-up** – *Frontend Fusion'24* by Datanerdz AI (Web Development Competition), July 2024